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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Clinic Prescription Record and Patient Visit Tracking System

Submitted by

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DECLARATION

I hereby declare that the project entitled "**Clinic Prescription Record and Patient Visit Tracking System"** submitted in partial fulfillment of the requirements for the course **B.Sc.Computer Science** is a **record of my original work** carried out under the guidance of **Mr. Dharaneesh** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Merlin M

CERTIFICATE

This is to certify that the project entitled **"Clinic Prescription Record and Patient Visit Tracking System"** is a **bona fide work** carried out by **Merlin M [1-'C'/2025-2026]**, **B.Sc.Computer Science, KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

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Submitted for the Viva-Voce Examination on. **10/04/2026**

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ABSTRACT

The Clinic Prescription Record and Patient Visit Tracking System is a Java-based application designed to manage patient records and prescriptions in small clinics. In many clinics, patient details and prescriptions are recorded manually in notebooks or files, which can lead to errors, loss of data, and difficulty in tracking patient visit history.

This project aims to develop a simple computerized system that allows clinic staff to record patient details, store prescriptions, and track patient visits efficiently. The system allows the user to enter patient information such as patient ID, name, age, disease, and prescription details. The program stores and displays the information in an organized manner.

The application is developed using Core Java programming concepts such as classes, objects, methods, loops, and conditional statements. The system helps reduce manual work, improve record management, and ensure better tracking of patient information.

This project demonstrates how Java can be used to create simple healthcare management systems for clinics.

1. INTRODUCTION

In healthcare facilities such as small clinics, maintaining patient records and prescriptions is an important task. Traditionally, many clinics maintain patient details manually using notebooks or paper records. This manual process often leads to several problems such as difficulty in finding past records, loss of data, and errors in maintaining prescription details.

With the advancement of computer technology, computerized systems can help improve the efficiency of patient record management. A digital system allows clinic staff to quickly enter patient details, store prescriptions, and retrieve patient visit information whenever required.

The Clinic Prescription Record and Patient Visit Tracking System is developed using the Java programming language to simplify the management of patient data in clinics. The system records patient details including patient ID, name, age, disease, and prescribed medicines.

1.1 Objective :

The main objective of the Clinic Prescription Record and Patient Visit Tracking System is to develop a simple Java application that manages patient prescriptions and visit records in clinics.

- The objectives of the system include:
- To develop a clinic management system using Java.
- To store patient information digitally.
- To record prescriptions given by doctors.
- To track patient visits efficiently.
- To reduce manual errors in maintaining records.
- To demonstrate the use of Java programming concepts.

1.2 Overview :

The Clinic Prescription Record and Patient Visit Tracking System is a simple console-based application developed in Java.

The system allows the user to enter the following details:

- Patient ID
- Patient Name
- Age
- Disease

Once the information is entered, the system displays the stored patient record. This helps the clinic staff maintain a digital record of patient visits and prescriptions.

The system demonstrates the use of Java features such as:

- Classes and Objects
- Scanner class for input
- Conditional statements
- Data storage using variables or arrays

1.3 Features :

The system includes the following features:

- **Patient Registration**

This module is used to enter and store basic details of the patient such as Patient ID, name, age, and contact information. It ensures that each patient is uniquely identified in the system for proper record management.

- **Disease Record**

This module is used to record the patient's illness or health problem during each visit. It stores details about the symptoms, diagnosis, or condition identified by the doctor.

- **Prescription Entry**

  This module allows the doctor to enter and store details of the medicines and treatments prescribed to the patient during each visit. It includes information such as medicine names, dosage, and duration.

- **Patient Visit Tracking**

  This module records each time a patient visits the clinic along with details such as visit date and purpose. It helps in maintaining a complete history of patient visits.

- **Simple User Interface**

  The system provides a clean and easy-to-use interface for clinic staff to operate without difficulty. It is designed in a simple manner so that users can quickly understand and navigate the system.

2. PROGRAM ANALYSIS

Problem Analysis is an important phase in system development where the existing problems are identified and studied in detail. It helps in understanding the need for a new system and defining the requirements clearly.

In many small clinics, patient records and prescriptions are still maintained manually using paper files or notebooks. This traditional method leads to several challenges and inefficiencies in managing patient information.

2.1 Problem Definition :

In many small clinics, patient information and prescriptions are written manually in notebooks or paper files. This manual method has several disadvantages:

- Patient records can be lost or damaged.
- Searching for old patient records is difficult.
- Maintaining visit history is time-consuming.
- Manual record keeping increases workload.

Therefore, a computerized system is required to maintain patient records efficiently.

The Clinic Prescription Record and Patient Visit Tracking System solves these problems by providing a digital platform to store and manage patient data.

2.2 Project Overview:

The system is a Java-based application that records and manages patient information in clinics.

- The system performs the following tasks:
- Accepts patient details from the user
- Records the disease and prescription
- Displays the patient record
- Helps track patient visits

This system improves the efficiency of clinic operations by providing a simple digital record system.

2.3 Scopes and Limitations:

Scope :

The system can be used in:

- Small clinics
- Medical centers
- Health camps
- Local hospitals

The system helps in managing patient records and prescriptions easily.

Limitation :

- The system is console-based.
- It stores only basic patient information.
- It does not use a database for long-term storage.

3. SYSTEM DESIGN

System Design is the phase where the structure and working of the system are planned before implementation. In this stage, the system is divided into different modules, and the flow of data is defined. A good design helps in developing the system efficiently and makes it easy to understand and maintain.

The Clinic Prescription Record and Patient Visit Tracking System is designed as a simple, user-friendly application where clinic staff can enter patient details, record prescriptions, and track visits.

3.1 Modules :

1. Patient Registration Module:

This module collects patient information such as:

- Patient ID
- Name
- Age

This data is stored in the system.

2. Disease Entry Module:

This module records the disease or medical condition of the patient.

Example:

- Fever
- Cold
- Headache
- Diabetes

3. Prescription Module:

This module stores the medicines prescribed by the doctor.

Example:

Paracetamol – 2 tablets daily

Antibiotics – 1 tablet after food

4. Record Display Module:

This module displays the stored patient information including:

- Patient ID
- Name
- Age
- Disease
- Prescription

5. Exit Module:

This module allows the user to exit the system safely.

4. METHODOLOGY

The development of the system followed these stages:

- Problem Analysis
- System Design
- Program Development
- Testing
- Result Documentation

4.1. Problem Analysis:

The need for a digital system for storing patient records was identified. The system requirements were analyzed to ensure that it meets clinic needs.

4.2. System Design:

System Design is the phase where the structure and working of the system are planned before implementation. In this stage, the system is divided into different modules, and the flow of data is defined. A good design helps in developing the system efficiently and makes it easy

4.3. Program Development:

The following concepts were used:

- Classes
- Objects
- Scanner class
- Conditional statements
- Methods

4.4. Testing and Validation:

The program was tested with different patient inputs to ensure correct functionality.

Input	Expected Result
Enter patient name	Name stored correctly
Enter disease	Disease displayed correctly
Enter prescription	Prescription recorded

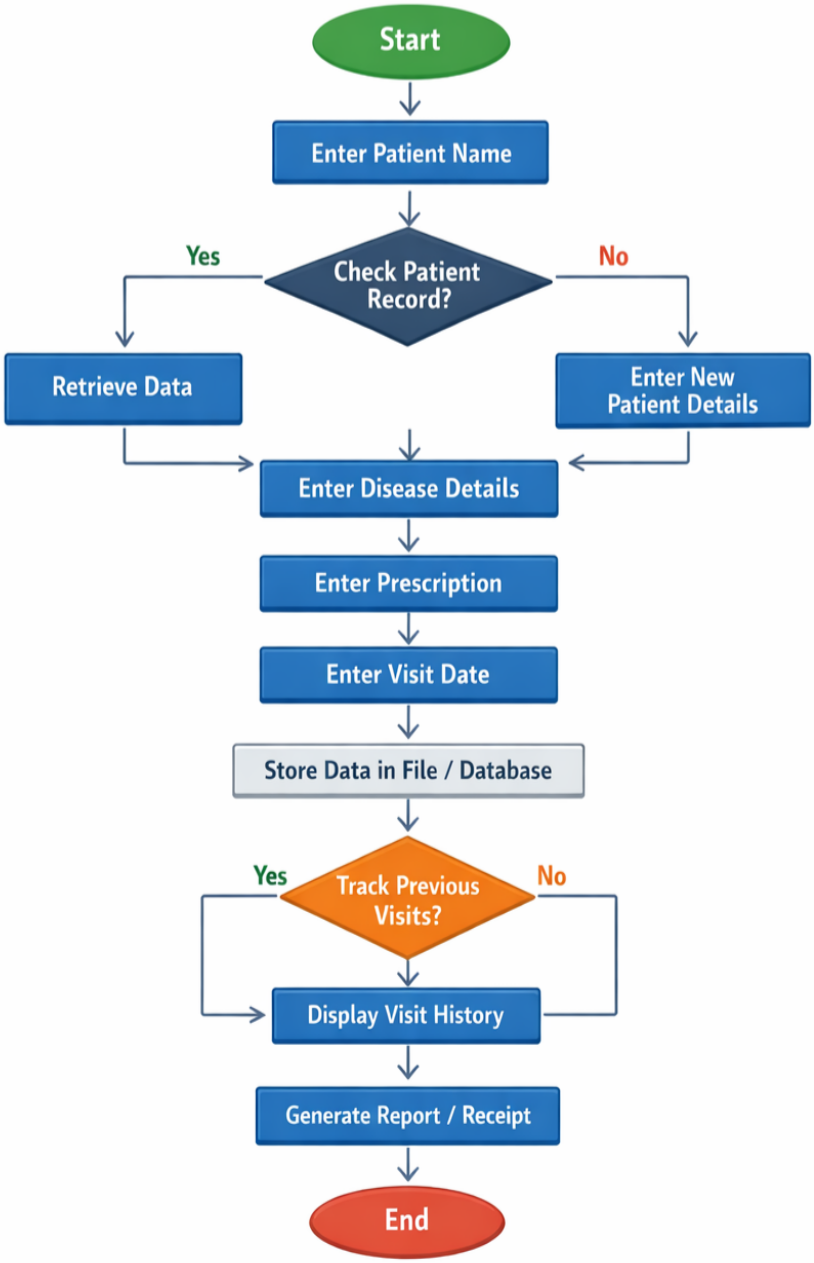
4.5. Result and Documentation:

The system successfully records patient information and displays the stored records. The results are shown through program output screens.

4.6. Algorithm

1. Start
2. Create Patient class (name, disease, date)
3. Create ArrayList to store patients
4. Display menu (Add, Search, Save, Exit)
5. Read user choice
6. If choice = 1
   Enter name, disease, date
   Store in ArrayList
7. If choice = 2
   Enter name
   Search in list
   If found, display details
   Else show “Not found”
8. If choice = 3
   Save all records into file
9. If choice = 4
   Exit program
10. End

4.7. Flowchart:



5. SOURCE CODE

```
import java.util.*;
import java.io.*;

public class Main {

   static class Patient {
      String name, disease, date;

      Patient(String n, String d, String dt) {
         name = n;
         disease = d;
         date = dt;
      }
   }

   public static void main(String[] args) throws Exception {

      Scanner sc = new Scanner(System.in);
      ArrayList<Patient> list = new ArrayList<>();

      int choice;

      do {
         System.out.println("\n1.Add Record  2.Search by Name  3.Save  4.Exit");
```

```
choice = sc.nextInt();
sc.nextLine();

// Add details
if (choice == 1) {
    System.out.print("Enter Name: ");
    String name = sc.nextLine();

    System.out.print("Enter Disease: ");
    String disease = sc.nextLine();

    System.out.print("Enter Date: ");
    String date = sc.nextLine();

    list.add(new Patient(name, disease, date));
    System.out.println("Record Saved!");
}

// Search by name
else if (choice == 2) {
    System.out.print("Enter Name: ");
    String name = sc.nextLine();

    boolean found = false;

    for (Patient p : list) {
        if (p.name.equalsIgnoreCase(name)) {
            System.out.println("\n--- Patient Details ---");
            System.out.println("Name: " + p.name);
```

```
                System.out.println("Disease: " + p.disease);
                System.out.println("Date: " + p.date);
                found = true;
            }
        }

        if (!found) {
            System.out.println("No Record Found!");
        }
    }

    // Save to file
    else if (choice == 3) {
        FileWriter fw = new FileWriter("clinic.txt");

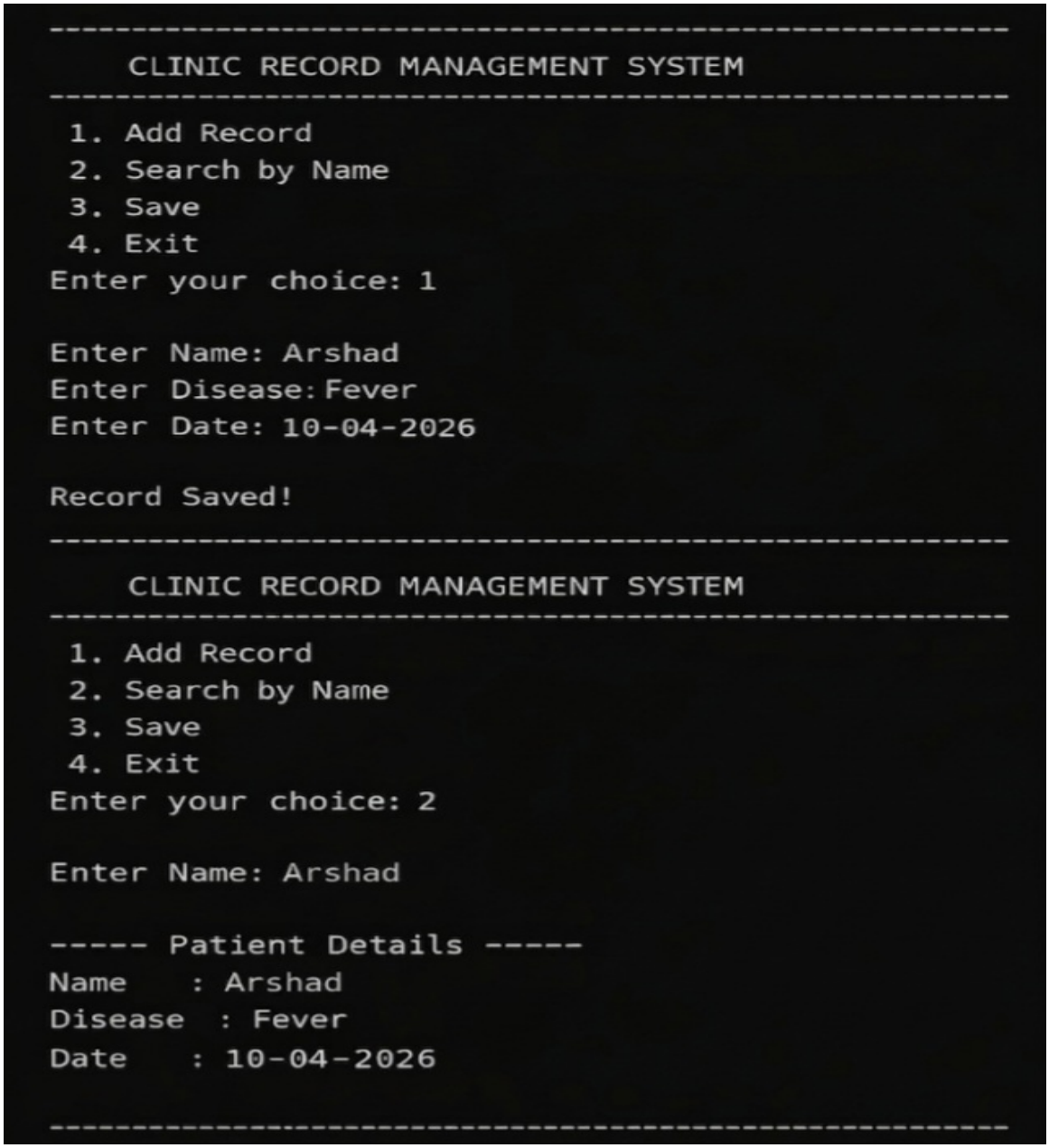
        for (Patient p : list) {
            fw.write(p.name + "," + p.disease + "," + p.date + "\n");
        }

        fw.close();
        System.out.println("Saved to file!");
    }

} while (choice != 4);

sc.close();
}
}
```

6. INPUT AND OUTPUT SCREENS



7. DISCUSSION & CONCLUSION

7.1 Discussion:

The Clinic Prescription Record and Patient Visit Tracking System helps clinics manage patient records digitally. The system reduces manual effort and improves record organization.

The application demonstrates the practical use of Java programming concepts in developing a healthcare management system.

7.2 Conclusion:

The project successfully implements a simple system for storing patient prescriptions and visit records. The system helps clinics maintain accurate patient records and improves efficiency.

- Future improvements may include:
- Database integration
- GUI interface
- Online patient records

8. REFERENCES

Books:

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- Core Java – Cay S. Horstmann

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